DEVELOPMENT OF A SMART IOT SYSTEM FOR PRECISION AGRICULTURE

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INTRODUCTION

- Traditional agriculture faces labor and monitoring challenges, affecting efficiency.
- Precision farming leverages sensors and data analytics for automation and better resource management.
- This project aims to create a precision farming system using sensor technologies and mesh networking.
- The goal is to enhance sustainability and food security through improved agricultural practices[1- 2].

NEED ANALYSIS

- Traditional farming results in inefficient resource use and higher crop failure risk.
- Precision farming leverages technology for optimized resource management and improved crop health.
- Real-time monitoring enables timely adjustments to environmental conditions for consistent growth.
- A unified dashboard enhances data analysis and reduces reliance on manual labor, promoting sustainability.

AIM, OBJECTIVES, NOVELTY & EXPECTED DELIVERABLES

Aim: To develop an IoT-based precision agriculture system for efficient data collection, automated control, and real-time monitoring.

Objectives:

1. To develop sensor nodes capable of collecting data efficiently and establish an ESP mesh network to facilitate communication among sensor nodes and control units.
2. To create server node capable of processing sensor data and executing commands for automated control.
3. To design a user-friendly dashboard for real-time monitoring and remote control of the sensor network.

Novelty:

- The solution eliminates the need for wired infrastructure, providing flexibility and efficiency in farming operations.
- Wireless connectivity allows real-time data collection, enabling quick decision-making and adjustments.
- An IoT-based approach enhances productivity and reduces costs while offering a sustainable solution for modern agriculture.

Expected Deliverables:

- Prototype sensor nodes and an ESP mesh network for efficient data collection and communication.
- A server node for processing sensor data and executing automated control commands.
- A user-friendly dashboard for real-time monitoring and remote control of the sensor network.

METHODOLOGY

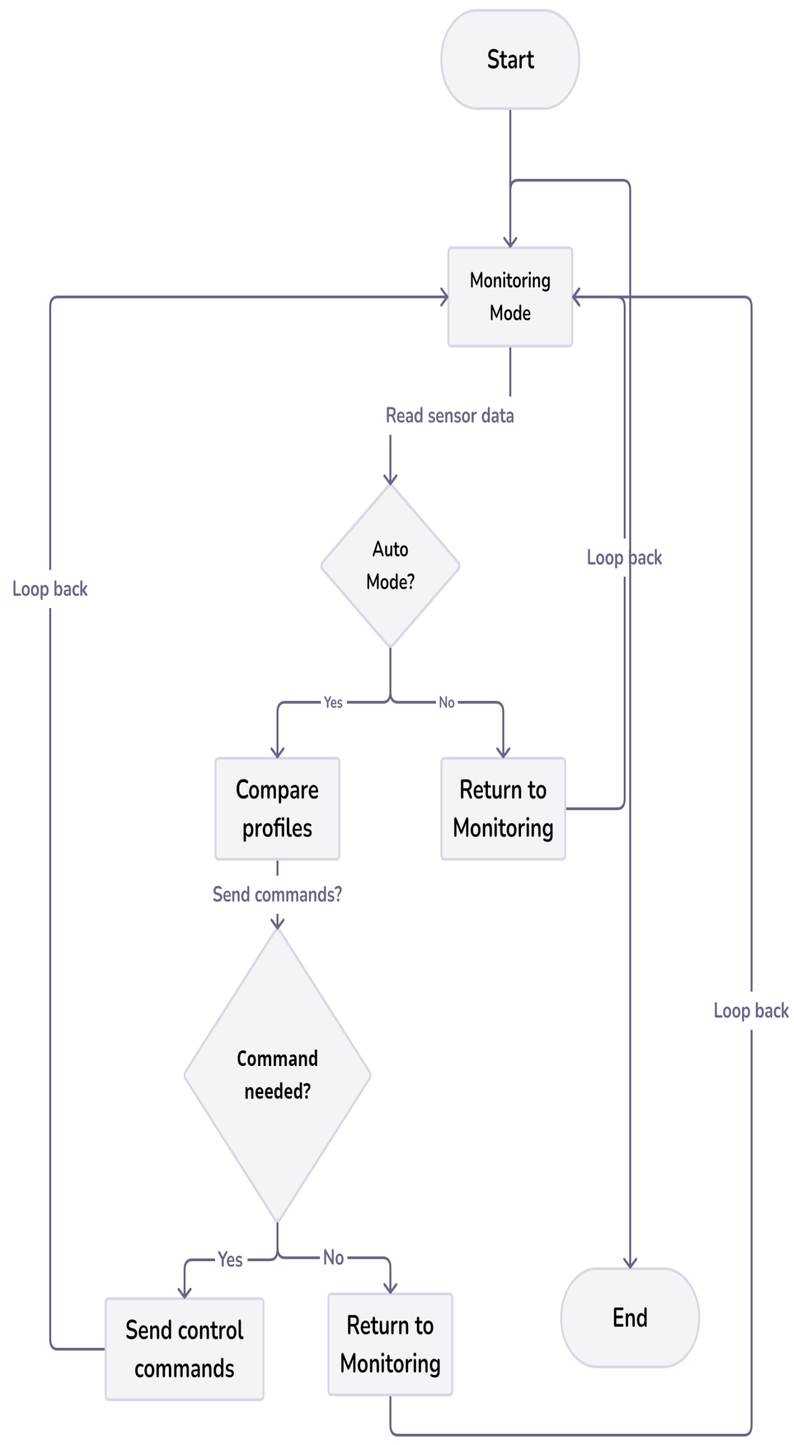

Figure 1: Flowchart of the Design Methodology for IOT System for Precision Agriculture

STANDARDS

Table 1: Standards for IOT System for Precision Agriculture

IEEE 802.11	This standard encompasses various specifications for medium access control (MAC) and physical layer (PHY) technologies, ensuring reliable data transfer and interoperability among wireless devices.[3].
IEEE 2700-2017	This standard defines key performance parameters for sensors, ensuring consistency and accuracy in data collection and reporting [4].

DESIGNING PARAMETERS

Table 2: Designing Parameters for IOT System for Precision Agriculture

Designing Parameters	Selection
Sensor	I2C Sensors , 6 Sensors
Communication	ESP-Now Protocol
Power Management	12V SMPS, Recharable Battery
Regulatory Compliance	IEEE2070-2017 (Standards)
Sample Testing	Jade crop plant

RESULTS AND DISCUSSION

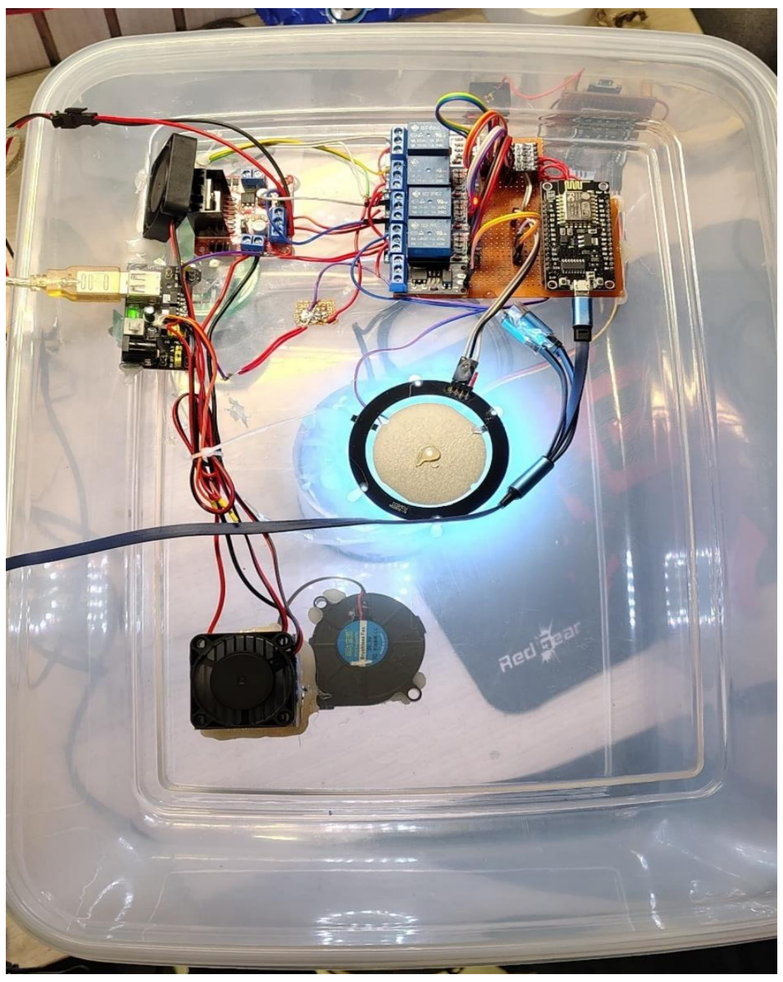

Figure 2: IOT System For Precision Agriculture

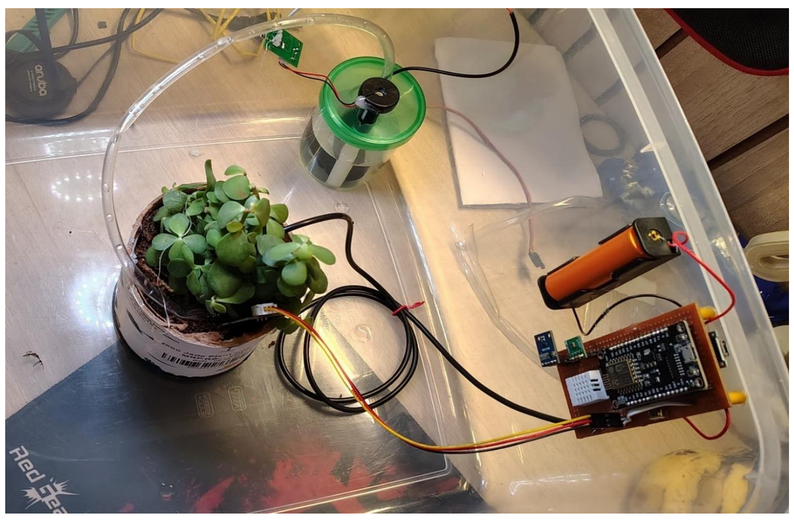

Figure 3: Closed Environment for the Crop

The prototype Hardware of the enclosure equipped with sensor and actuator network along with the sample crop plant is shown in Fig 2 and Fig 3.

CONCLUSIONS

- The project integrates sensor nodes and an ESP mesh network for reliable real-time data communication.
- A server node optimises automation, enabling real-time processing and quick responses to data inputs.
- A user-friendly dashboard improves accessibility, allowing intuitive monitoring and management of the system.

REFERENCES

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